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ASPECTS OF NON-ABELIAN GROUP-BASED CRYPTOGRAPHY: A NEW APPROACH TO MODERN CRYPTOGRAPHY

DR. DELARAM KAHROBAEI

ABSTRACT. In this course, we cover different aspects of non-abelian group-based cryptography. This topic, which is about a decade old, originally was initiated by two group theorists and a number theorist Anshel-Anshel-Goldfeld. In this course, I will cover the basic definitions and facts about group theory (computational, combinatorial), discuss the decision and search problems and their complexities (computational and generic case). I then bring applications of these abstract structures to non-abelian public key cryptography, digital signatures, secret sharing, and bit commitments and suggested platform groups. At the same time I will cover the mathematical definitions of provably secure cryptosystems in this context. I also plan to analyze the security of each discussed system and study the possible attacks. The course will be of interest of computer science and mathematics students in every level of their studies who are interested to see the future of cryptography beyond the traditional platforms and number theoretic approaches. Lecture notes and slides will be distributed during the class, no prior knowledge of programming, group theory, cryptography is required. The grade is based on class participation and a project at the end of the semester.

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Date: November 7, 2010.